Homework I — Solutions

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Contents

Problem 1 **2**
1.A) . . . 2
1.B) . . . 2

Problem 2 **3**
2.A) . . . 3
2.B) . . . 3

Problem 3 **5**

Problem 4 **7**
4.A) . . . 7
4.B) . . . 7
4.C) . . . 7

Problem 5 **9**

A Conventions **10**
A.A) Metric . . . 10
A.B) Levi-Civita . . . 10

B Determinants **11**

C Noether's Theorem **12**

D Noether's Second Theorem **15**

E Boundary Conditions **17**

F Algebra of conserved quantities **20**
F.A) Internal Symmetries . . . 20
F.B) Space-Time Symmetries . . . 21

Problem 1

1.A)

1.B)

Problem 2

2.A)

$$0 = \frac{\delta}{\delta A_\alpha(y)} S[A_\mu] \tag{2.1}$$

$$0 = -\frac{1}{4}\frac{\delta}{\delta A_\alpha(y)} \int \mathrm{d}^4x\, F^{\mu\nu} F_{\mu\nu} \tag{2.2}$$

$$0 = -\frac{1}{2} \int \mathrm{d}^4x\, F^{\mu\nu} \frac{\delta}{\delta A_\alpha(y)} F_{\mu\nu} \tag{2.3}$$

$$0 = -\frac{1}{2} \int \mathrm{d}^4x\, F^{\mu\nu} \big(\partial_\mu g_\nu{}^\alpha \delta^{(4)}(x-y) - \partial_\nu g_\mu{}^\alpha \delta^{(4)}(x-y)\big) \tag{2.4}$$

$$0 = \int \mathrm{d}^4x\, \partial_\mu F^{\mu\nu} g_\nu{}^\alpha \delta^{(4)}(x-y) \tag{2.5}$$

$$0 = \partial_\mu F^{\mu\alpha} \tag{2.6}$$

2.B)

By eq. (C.21),

$$J_I^\mu = \frac{\partial \mathcal{L}}{\partial \partial_\mu \phi(x)} \frac{\mathrm{d}\phi'(x)}{\mathrm{d}a^I}\bigg|_{a^I=0} + \mathcal{L}\frac{\mathrm{d}x'^\mu}{\mathrm{d}a^I}\bigg|_{a^I=0} \tag{2.7}$$

In this case the transformation is,

$$\begin{cases} x & \to x'^\mu = x^\mu + a^\mu \\ A_\alpha(x) & \to A'_\alpha(x') = A_\alpha(x) \end{cases} \Rightarrow \frac{\mathrm{d}x'^\mu}{\mathrm{d}a^\nu}\bigg|_{a^\nu=0} = g^\mu{}_\nu, \quad \frac{\mathrm{d}A'_\alpha(x)}{\mathrm{d}a^\nu}\bigg|_{a^\nu=0} = -\partial_\nu A_\alpha(x) \tag{2.8}$$

Thus,

$$J^\mu{}_\nu = \frac{\partial \mathcal{L}}{\partial \partial_\mu A_\alpha(x)} \frac{\mathrm{d}A'_\alpha(x)}{\mathrm{d}a^\nu}\bigg|_{a^\nu=0} + \mathcal{L}\frac{\mathrm{d}x'^\mu}{\mathrm{d}a^\nu}\bigg|_{a^\nu=0} \tag{2.9}$$

$$J^\mu{}_\nu = -\frac{\partial \mathcal{L}}{\partial \partial_\mu A_\alpha(x)} \partial_\nu A_\alpha(x) + \mathcal{L} g^\mu{}_\nu \tag{2.10}$$

$$J^\mu{}_\nu = \frac{1}{2} F^{\rho\sigma} \frac{\partial F_{\rho\sigma}}{\partial \partial_\mu A_\alpha(x)} \partial_\nu A_\alpha(x) - \frac{1}{4} F_{\rho\sigma} F^{\rho\sigma} g^\mu{}_\nu \tag{2.11}$$

$$J^\mu{}_\nu = \frac{1}{2} F^{\rho\sigma} \big(g_\rho{}^\mu g_\sigma{}^\alpha - g_\rho{}^\alpha g_\sigma{}^\mu\big) \partial_\nu A_\alpha(x) - \frac{1}{4} F_{\rho\sigma} F^{\rho\sigma} g^\mu{}_\nu \tag{2.12}$$

$$J^\mu{}_\nu = \frac{1}{2} F^{\rho\sigma} \big(\partial_\nu g_\rho{}^\mu A_\sigma - \partial_\nu A_\rho g_\sigma{}^\mu\big) - \frac{1}{4} F_{\rho\sigma} F^{\rho\sigma} g^\mu{}_\nu \tag{2.13}$$

$$J^\mu{}_\nu = F^{\mu\sigma} \partial_\nu A_\sigma - \frac{1}{4} F_{\rho\sigma} F^{\rho\sigma} g^\mu{}_\nu \tag{2.14}$$

Let's check that this quantity is conserved when the equations of motion are obeyed,

$$\partial_\mu J^\mu{}_\nu = \partial_\mu F^{\mu\sigma} \partial_\nu A_\sigma + F^{\mu\sigma} \partial_\mu \partial_\nu A_\sigma - \frac{1}{2} F^{\rho\sigma} \partial_\mu F_{\rho\sigma} g^\mu{}_\nu \tag{2.15}$$

$$\partial_\mu J^\mu{}_\nu = \partial_\mu F^{\mu\sigma}\partial_\nu A_\sigma + F^{\mu\sigma}\partial_\mu\partial_\nu A_\sigma - F^{\rho\sigma}\partial_\nu\partial_\rho A_\sigma = 0 \tag{2.16}$$

Yes! But, it isn't gauge invariant, neither symmetric. How to obtain a gauge invariant and symmetric energy momentum tensor?

Problem 3

The transformation is,

$$\begin{cases} x & \to x' = \mathrm{e}^a x \\ \phi(x) & \to \phi'(x') = \mathrm{e}^{-d_\phi a}\phi(x) \end{cases} \Rightarrow \left.\frac{\mathrm{d}x'^\mu}{\mathrm{d}a}\right|_{a=0} = x^\mu, \quad \left.\frac{\mathrm{d}\phi'(x)}{\mathrm{d}a}\right|_{a=0} = -d_\phi\phi(x) - \partial_\mu\phi(x)x^\mu \tag{3.1}$$

First let's check that it's a symmetry,

$$S'_a[\phi'] = \int \mathrm{d}^4x' \left[-\frac{1}{2}\partial'_\mu\phi'(x')\partial'^\mu\phi'(x') - \frac{\lambda}{4!}\phi'^4(x')\right] \tag{3.2}$$

$$S'_a[\phi'] = \int \mathrm{d}^4x\, \mathrm{e}^{4a}\left[-\frac{1}{2}\mathrm{e}^{-2a-2d_\phi a}\partial_\mu\phi(x)\partial^\mu\phi(x) - \mathrm{e}^{-4d_\phi a}\frac{\lambda}{4!}\phi^4(x)\right] \tag{3.3}$$

$$S'_a[\phi'] = \int \mathrm{d}^4x \left[-\frac{1}{2}\mathrm{e}^{2a-2d_\phi a}\partial_\mu\phi(x)\partial^\mu\phi(x) - \mathrm{e}^{4a-4d_\phi a}\frac{\lambda}{4!}\phi^4(x)\right] \tag{3.4}$$

For $d_\phi = 1$,

$$S'_a[\phi'] = \int \mathrm{d}^4x \left[-\frac{1}{2}\partial_\mu\phi(x)\partial^\mu\phi(x) - \frac{\lambda}{4!}\phi^4(x)\right] = S[\phi] \tag{3.5}$$

Now, let's compute the conserved charge,

$$J^\mu = \frac{\partial\mathcal{L}}{\partial\partial_\mu\phi(x)}\left.\frac{\mathrm{d}\phi'(x)}{\mathrm{d}a}\right|_{a=0} + \mathcal{L}\left.\frac{\mathrm{d}x'^\mu}{\mathrm{d}a}\right|_{a=0} \tag{3.6}$$

$$J^\mu = -\frac{\partial\mathcal{L}}{\partial\partial_\mu\phi(x)}(d_\phi\phi(x) + \partial_\alpha\phi(x)x^\alpha) + \mathcal{L}x^\mu \tag{3.7}$$

$$J^\mu = \partial^\mu\phi(d_\phi\phi + \partial_\alpha\phi x^\alpha) + \left(-\frac{1}{2}\partial^\alpha\phi\partial_\alpha\phi + \frac{\lambda}{4!}\phi^4\right)x^\mu \tag{3.8}$$

Check if it is really divergenceless,

$$\begin{aligned}\partial_\mu J^\mu = {} & \partial_\mu\partial^\mu\phi(d_\phi\phi + \partial_\alpha\phi x^\alpha) + \partial^\mu\phi(d_\phi\partial_\mu\phi + \partial_\mu\partial_\alpha\phi x^\alpha + \partial_\alpha\phi\partial_\mu x^\alpha) \\ & + \left(-\partial^\alpha\phi\partial_\mu\partial_\alpha\phi - \frac{\lambda}{3!}\phi^3\partial_\mu\phi\right)x^\mu + \left(-\frac{1}{2}\partial^\alpha\phi\partial_\alpha\phi - \frac{\lambda}{4!}\phi^4\right)\partial_\mu x^\mu\end{aligned} \tag{3.9}$$

$$\begin{aligned}\partial_\mu J^\mu = {} & \Box\phi(d_\phi\phi + \partial_\alpha\phi x^\alpha) + \partial^\mu\phi\big(d_\phi\partial_\mu\phi + \partial_\mu\partial_\alpha\phi x^\alpha + \partial_\alpha\phi g_\mu{}^\alpha\big) \\ & + \left(-\partial^\alpha\phi\partial_\mu\partial_\alpha\phi - \frac{\lambda}{3!}\phi^3\partial_\mu\phi\right)x^\mu + 4\left(-\frac{1}{2}\partial^\alpha\phi\partial_\alpha\phi - \frac{\lambda}{4!}\phi^4\right)\end{aligned} \tag{3.10}$$

$$\begin{aligned}\partial_\mu J^\mu = {} & \left(\Box\phi - \frac{\lambda}{3!}\phi^3\right)x^\alpha\partial_\alpha\phi + \left(d_\phi\Box\phi - \frac{\lambda}{3!}\phi^3\right)\phi + \partial^\mu\phi(d_\phi\partial_\mu\phi + \partial_\mu\partial_\alpha\phi x^\alpha + \partial_\mu\phi) \\ & - \partial^\alpha\phi\partial_\mu\partial_\alpha\phi x^\mu - 2\partial^\alpha\phi\partial_\alpha\phi\end{aligned} \tag{3.11}$$

$$\partial_\mu J^\mu = \left(\Box\phi - \frac{\lambda}{3!}\phi^3\right)x^\alpha\partial_\alpha\phi + \left(\Box\phi - \frac{\lambda}{3!}\phi^3\right)\phi + (d_\phi - 1)\phi\Box\phi + (d_\phi - 1)\partial^\mu\phi\partial_\mu\phi \tag{3.12}$$

Using the equation of motion,

$$\Box\phi = \frac{\lambda}{3!}\phi^3 \tag{3.13}$$

We get,

$$\partial_\mu J^\mu = (d_\phi - 1)\phi\Box\phi + (d_\phi - 1)\partial^\mu\phi\partial_\mu\phi \tag{3.14}$$

Now set $d_\phi = 1$,

$$\partial_\mu J^\mu = 0 \tag{3.15}$$

Problem 4

4.A)

The Lorentz transformation is,

$$x'^{\mu} = \Lambda^{\mu}{}_{\nu} x^{\nu} = x^{\mu} + \omega^{\mu}{}_{\nu} x^{\nu} + \mathcal{O}\big(\omega^2\big) \tag{4.1}$$

Which has to satisfy,

$$\begin{aligned}
y^{\mu} x_{\mu} &= y'^{\mu} x'_{\mu} && (4.2)\\
y^{\mu} x_{\mu} &= \big(y^{\mu} + \omega^{\mu}{}_{\nu} y^{\nu} + \mathcal{O}\big(\omega^2\big)\big)\big(x_{\mu} + \omega_{\mu\alpha} x^{\alpha} + \mathcal{O}\big(\omega^2\big)\big) && (4.3)\\
y^{\mu} x_{\mu} &= y^{\mu} x_{\mu} + \omega^{\mu}{}_{\nu} y^{\nu} x_{\mu} + y^{\mu} \omega_{\mu\alpha} x^{\alpha} + \mathcal{O}\big(\omega^2\big) && (4.4)\\
0 &= \omega_{\mu\nu} y^{\nu} x^{\mu} + \omega_{\mu\nu} x^{\nu} y^{\mu} && (4.5)\\
0 &= y^{\nu} x^{\mu} \big(\omega_{\mu\nu} + \omega_{\nu\mu}\big) \Rightarrow \omega_{\mu\nu} + \omega_{\nu\mu} = 0 \Rightarrow \omega_{\mu\nu} = -\omega_{\nu\mu} && (4.6)
\end{aligned}$$

4.B)

A rotation should preserve the time component,

$$\begin{aligned}
x^0 = x'^0 &= x^0 + \omega^0{}_{\nu} x^{\nu} && (4.7)\\
0 &= \omega^0{}_{\nu} x^{\nu} \Rightarrow \omega^0{}_{\nu} = 0 \Rightarrow \omega_{0\nu} = 0 && (4.8)
\end{aligned}$$

Together with the anti-symmetry gives,

$$\omega_{\mu\nu} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & \omega_{12} & -\omega_{31} \\ 0 & -\omega_{12} & 0 & \omega_{23} \\ 0 & \omega_{31} & -\omega_{23} & 0 \end{pmatrix} \tag{4.9}$$

4.C)

The Lorentz transformation for a scalar field is,

$$\begin{cases} x^{\mu} & \to x'^{\mu} = \Lambda^{\mu}{}_{\nu}(\omega) x^{\nu} \\ \phi(x) & \to \phi'(x') = \phi(x) \end{cases} \tag{4.10}$$

Which is parametrized by $\omega_{\mu\nu}$, thus,

$$\begin{aligned}
\left.\frac{\mathrm{d}x'^{\mu}}{\mathrm{d}\omega^{\alpha\beta}}\right|_{\omega_{\alpha\beta}=0} &= \left.\frac{\mathrm{d}}{\mathrm{d}\omega_{\alpha\beta}}\big(x^{\mu} + \omega^{\mu}{}_{\nu} x^{\nu} + \mathcal{O}\big(\omega^2\big)\big)\right|_{\omega_{\alpha\beta}=0} && (4.11)\\
\left.\frac{\mathrm{d}x'^{\mu}}{\mathrm{d}\omega^{\alpha\beta}}\right|_{\omega_{\alpha\beta}=0} &= \big(g_{\alpha}{}^{\mu} g_{\beta\nu} - g_{\beta}{}^{\mu} g_{\alpha\nu}\big) x^{\nu} = g_{\alpha}{}^{\mu} x_{\beta} - g_{\beta}{}^{\mu} x_{\alpha} && (4.12)
\end{aligned}$$

And,

$$\frac{\mathrm{d}\phi'(x)}{\mathrm{d}\omega^{\alpha\beta}}\bigg|_{\omega^{\alpha\beta}=0} = \frac{\mathrm{d}}{\mathrm{d}\omega^{\alpha\beta}}\big(\phi\big(\Lambda^{-1}x\big)\big)\bigg|_{\omega^{\alpha\beta}=0} = \partial_\nu\phi(x)\frac{\mathrm{d}}{\mathrm{d}\omega^{\alpha\beta}}\big(\Lambda^{-1}x\big)^\nu\bigg|_{\omega^{\alpha\beta}=0} \tag{4.13}$$

$$\frac{\mathrm{d}\phi'(x)}{\mathrm{d}\omega^{\alpha\beta}}\bigg|_{\omega^{\alpha\beta}=0} = \partial_\nu\phi(x)\frac{\mathrm{d}\Lambda_\rho{}^\nu}{\mathrm{d}\omega^{\alpha\beta}}\bigg|_{\omega^{\alpha\beta}=0} x^\rho = \partial_\nu\phi(x)\big(g_{\alpha\rho}g_\beta{}^\nu - g_{\beta\rho}g_\alpha{}^\nu\big)x^\rho \tag{4.14}$$

$$\frac{\mathrm{d}\phi'(x)}{\mathrm{d}\omega^{\alpha\beta}}\bigg|_{\omega^{\alpha\beta}=0} = x_\alpha\partial_\beta\phi - x_\beta\partial_\alpha\phi \tag{4.15}$$

Using these results in eq. (C.21),

$$J^\mu{}_{\alpha\beta} = \frac{\partial\mathcal{L}}{\partial\partial_\mu\phi(x)}\frac{\mathrm{d}\phi'(x)}{\mathrm{d}\omega^{\alpha\beta}}\bigg|_{\omega^{\alpha\beta}=0} + \mathcal{L}\frac{\mathrm{d}x'^\mu}{\mathrm{d}\omega^{\alpha\beta}}\bigg|_{\omega^{\alpha\beta}=0} \tag{4.16}$$

$$J^\mu{}_{\alpha\beta} = \frac{\partial\mathcal{L}}{\partial\partial_\mu\phi(x)}(x_\alpha\partial_\beta\phi - x_\beta\partial_\alpha\phi) + \mathcal{L}\big(g_\alpha{}^\mu x_\beta - g_\beta{}^\mu x_\alpha\big) \tag{4.17}$$

$$J^\mu{}_{\alpha\beta} = x_\beta J^\mu{}_\alpha - x_\alpha J^\mu{}_\beta \tag{4.18}$$

Check that is conserved,

$$\partial_\mu J^\mu{}_{\alpha\beta} = \partial_\mu\big(x_\beta J^\mu{}_\alpha - x_\alpha J^\mu{}_\beta\big) \tag{4.19}$$

$$\partial_\mu J^\mu{}_{\alpha\beta} = \partial_\mu x_\beta J^\mu{}_\alpha + x_\beta\partial_\mu J^\mu{}_\alpha - \partial_\mu x_\alpha J^\mu{}_\beta - x_\alpha\partial_\mu J^\mu{}_\beta \tag{4.20}$$

$$\partial_\mu J^\mu{}_{\alpha\beta} = g_{\mu\beta}J^\mu{}_\alpha + x_\beta\partial_\mu J^\mu{}_\alpha - g_{\mu\alpha}J^\mu{}_\beta - x_\alpha\partial_\mu J^\mu{}_\beta \tag{4.21}$$

$$\partial_\mu J^\mu{}_{\alpha\beta} = J_{\beta\alpha} - J_{\alpha\beta} + x_\beta\partial_\mu J^\mu{}_\alpha - x_\alpha\partial_\mu J^\mu{}_\beta \tag{4.22}$$

Without a proper lagrangian this is as far as we can go, but, an interesting result that can be seen from here is, whenever the energy momentum tensor is conserved and symmetric, the angular momentum tensor is conserved.

Problem 5

The Lorentz transformation for a vector field is,

$$\begin{cases} x^\mu & \to x'^\mu = \Lambda^\mu{}_\nu(\omega)x^\nu \\ A^\rho(x) & \to A'^\rho(x') = \Lambda^\rho{}_\sigma A^\sigma(x) \end{cases} \tag{5.1}$$

The first step we had already made, eq. (4.12),

$$\left.\frac{\mathrm{d}x'^\mu}{\mathrm{d}\omega^{\alpha\beta}}\right|_{\omega_{\alpha\beta}=0} = g_\alpha{}^\mu x_\beta - g_\beta{}^\mu x_\alpha \tag{5.2}$$

The second step,

$$\left.\frac{\mathrm{d}A'^\rho(x)}{\mathrm{d}\omega^{\alpha\beta}}\right|_{\omega^{\alpha\beta}=0} = \left.\frac{\mathrm{d}}{\mathrm{d}\omega^{\alpha\beta}}\big(\Lambda^\rho{}_\sigma A^\sigma\big(\Lambda^{-1}x\big)\big)\right|_{\omega^{\alpha\beta}=0} \tag{5.3}$$

$$\left.\frac{\mathrm{d}A'^\rho(x)}{\mathrm{d}\omega^{\alpha\beta}}\right|_{\omega^{\alpha\beta}=0} = \left.\frac{\mathrm{d}}{\mathrm{d}\omega^{\alpha\beta}}(\Lambda^\rho{}_\sigma)\right|_{\omega^{\alpha\beta}=0} A^\sigma(x) + g^\rho{}_\sigma \left.\frac{\mathrm{d}}{\mathrm{d}\omega^{\alpha\beta}}\big(A^\sigma\big(\Lambda^{-1}x\big)\big)\right|_{\omega^{\alpha\beta}=0} \tag{5.4}$$

$$\left.\frac{\mathrm{d}A'^\rho(x)}{\mathrm{d}\omega^{\alpha\beta}}\right|_{\omega^{\alpha\beta}=0} = \big(g_\alpha{}^\rho g_{\beta\sigma} - g_\beta{}^\rho g_{\alpha\sigma}\big)A^\sigma(x) + \partial_\nu A^\rho(x)\left.\frac{\mathrm{d}}{\mathrm{d}\omega^{\alpha\beta}}\big(\Lambda^{-1}x\big)^\nu\right|_{\omega^{\alpha\beta}=0} \tag{5.5}$$

$$\left.\frac{\mathrm{d}A'^\rho(x)}{\mathrm{d}\omega^{\alpha\beta}}\right|_{\omega^{\alpha\beta}=0} = g_\alpha{}^\rho A_\beta - g_\beta{}^\rho A_\alpha + \partial_\nu A^\rho(x)\left.\frac{\mathrm{d}\Lambda_\gamma{}^\nu}{\mathrm{d}\omega^{\alpha\beta}}\right|_{\omega^{\alpha\beta}=0} x^\gamma \tag{5.6}$$

$$\left.\frac{\mathrm{d}A'^\rho(x)}{\mathrm{d}\omega^{\alpha\beta}}\right|_{\omega^{\alpha\beta}=0} = g_\alpha{}^\rho A_\beta - g_\beta{}^\rho A_\alpha + \partial_\nu A^\rho(x)\big(g_{\alpha\gamma}g_\beta{}^\nu - g_{\beta\gamma}g_\alpha{}^\nu\big)x^\gamma \tag{5.7}$$

$$\left.\frac{\mathrm{d}A'^\rho(x)}{\mathrm{d}\omega^{\alpha\beta}}\right|_{\omega^{\alpha\beta}=0} = g_\alpha{}^\rho A_\beta - g_\beta{}^\rho A_\alpha + x_\alpha\partial_\beta A^\rho(x) - x_\beta\partial_\alpha A^\rho(x) \tag{5.8}$$

$$\tag{5.9}$$

Using these results in eq. (C.21),

$$J^\mu{}_{\alpha\beta} = \frac{\partial\mathcal{L}}{\partial\partial_\mu A^\rho(x)}\left.\frac{\mathrm{d}A'^\rho(x)}{\mathrm{d}\omega^{\alpha\beta}}\right|_{\omega^{\alpha\beta}=0} + \mathcal{L}\left.\frac{\mathrm{d}x'^\mu}{\mathrm{d}\omega^{\alpha\beta}}\right|_{\omega^{\alpha\beta}=0} \tag{5.10}$$

$$J^\mu{}_{\alpha\beta} = \frac{\partial\mathcal{L}}{\partial\partial_\mu A^\rho(x)}\big(g_\alpha{}^\rho A_\beta - g_\beta{}^\rho A_\alpha + x_\alpha\partial_\beta A^\rho - x_\beta\partial_\alpha A^\rho\big) + \mathcal{L}\big(g_\alpha{}^\mu x_\beta - g_\beta{}^\mu x_\alpha\big) \tag{5.11}$$

$$J^\mu{}_{\alpha\beta} = x_\beta J^\mu{}_\alpha - x_\alpha J^\mu{}_\beta + \frac{\partial\mathcal{L}}{\partial\partial_\mu A^\alpha(x)}A_\beta - \frac{\partial\mathcal{L}}{\partial\partial_\mu A^\beta(x)}A_\alpha \tag{5.12}$$

Which also isn't gauge invariant. To solve this problem consider the following,

$$J^\mu{}_{\alpha\beta} = x_\beta J^\mu{}_\alpha - x_\alpha J^\mu{}_\beta + S^\mu{}_{\alpha\beta} \tag{5.13}$$

The conservation equation gives,

$$\partial_\mu J^\mu{}_{\alpha\beta} = J_{\beta\alpha} - J_{\alpha\beta} + \partial_\mu S^\mu{}_{\alpha\beta} = 0 \tag{5.14}$$

$$\partial_\mu S^\mu{}_{\alpha\beta} = J_{\alpha\beta} - J_{\beta\alpha} \tag{5.15}$$

Thus, we can get from $J_{\alpha\beta}$ a symmetric tensor by summing a divergenceless tensor,

$$\Theta_{\alpha\beta} = J_{\alpha\beta} + \frac{1}{2}\big(J_{\alpha\beta} - J_{\beta\alpha}\big) - \frac{1}{2}\partial\big(J_{\alpha\beta} - J_{\beta\alpha}\big) = J_{\alpha\beta} + \frac{1}{2}\partial_\mu S^\mu{}_{\alpha\beta} \tag{5.16}$$

A Conventions

A.A) Metric

There are two choices of metric,

$$g_- = \mathrm{diag}\big(- \quad + \quad + \quad +\big), \quad g_+ = \mathrm{diag}\big(+ \quad - \quad - \quad -\big) \tag{A.1}$$

I will always use g_-. This changes a lot of signs in a lot of places, as example,

$$\mathcal{L} = g_+^{\mu\nu}\frac{1}{2}\partial_\mu\phi\partial_\nu\phi - V[\phi] = -g_-^{\mu\nu}\frac{1}{2}\partial_\mu\phi\partial_\nu\phi - V[\phi] \tag{A.2}$$

Physics should be invariant under conventions, so they don't matter, as long as you know which conventions you are using and are consistent.

A.B) Levi-Civita

The Levi-Civita is the (unique) totally antisymmetric four-pseudo-tensor, $\epsilon_{\alpha\beta\mu\nu}$, there are two possible choices of normalization,

$$\epsilon^+_{0123} = 1, \quad \epsilon^-_{0123} = -1 \tag{A.3}$$

I will always use ϵ^+.

B Determinants

C Noether's Theorem

Here we're going to derive the Noether theorem for a handful of cases, first of all, the validity of this theorem is to global continuous symmetries, which in our field theory context means: we start of with an action which depends upon a field, $S[\phi]$, it's supposed to encode whatever index the fields have, then, a symmetry transformation is given in terms of:

$$\begin{cases} x^\mu & \to x'^\mu = x'^\mu(x;a) \\ \phi(x) & \to \phi'(x') = \phi'(\phi(x), x; a) \end{cases} \Rightarrow S[\phi] = S'_a[\phi'] \tag{C.1}$$

Let's look further at the symmetry condition,

$$S'_a[\phi'] - S[\phi] = \int_{\Omega'} \mathrm{d}^4x' \, \mathcal{L}'[\phi'(x'); x'] - \int_{\Omega} \mathrm{d}^4x \, \mathcal{L}[\phi(x); x] \tag{C.2}$$

$$\begin{aligned} S'_a[\phi'] - S[\phi] = & \int_{\Omega} \mathrm{d}^4x \, \mathcal{L}'[\phi'(x); x] - \int_{\Omega} \mathrm{d}^4x \, \mathcal{L}[\phi(x); x] \\ & + \int_{\Omega'} \mathrm{d}^4x' \, \mathcal{L}'[\phi'(x'); x'] - \int_{\Omega} \mathrm{d}^4x \, \mathcal{L}'[\phi'(x); x] \end{aligned} \tag{C.3}$$

$$\begin{aligned} S'_a[\phi'] - S[\phi] = & \int_{\Omega} \mathrm{d}^4x \, \{\mathcal{L}'[\phi'(x); x] - \mathcal{L}[\phi(x); x]\} \\ & + \int_{\Omega} \mathrm{d}^4x \left\{ \mathrm{Det}\left[\frac{\partial x'}{\partial x}\right] \mathcal{L}'[\phi'(x'); x'] - \mathcal{L}'[\phi'(x); x] \right\} \end{aligned} \tag{C.4}$$

$$\begin{aligned} S'_a[\phi'] - S[\phi] = & \int_{\Omega} \mathrm{d}^4x \, \{\mathcal{L}'[\phi'(x); x] - \mathcal{L}[\phi(x); x]\} \\ & + \int_{\Omega} \mathrm{d}^4x \left\{ \mathrm{Det}\left[\frac{\partial x'}{\partial x}\right] - 1 \right\} \mathcal{L}'[\phi'(x'); x'] \\ & + \int_{\Omega} \mathrm{d}^4x \, \{\mathcal{L}'[\phi'(x'); x'] - \mathcal{L}'[\phi'(x); x]\} \end{aligned} \tag{C.5}$$

Suppose $\mathcal{L}' = \mathcal{L}$,

$$\begin{aligned} \left.\frac{[S'_a[\phi'] - S[\phi]]}{a}\right|_{a\to 0} = & \int_{\Omega} \mathrm{d}^4x \int_{\Omega} \mathrm{d}^4y \left\{ \frac{\delta \mathcal{L}}{\delta \phi(y)} \frac{\phi'(y) - \phi(y)}{a} \right\}\Bigg|_{a\to 0} \\ & + \int_{\Omega} \mathrm{d}^4x \, \frac{1}{a} \left\{ \mathrm{Det}\left[\frac{\partial x'}{\partial x}\right] - 1 \right\}\Bigg|_{a\to 0} \mathcal{L} \\ & + \int_{\Omega} \mathrm{d}^4x \, \frac{\mathrm{d}\mathcal{L}}{\mathrm{d}x^\alpha} \frac{x'^\alpha - x^\alpha}{a}\Bigg|_{a\to 0} \end{aligned}$$

$$\begin{aligned} \left.\frac{\mathrm{d}}{\mathrm{d}a} S'_a[\phi']\right|_{a=0} = & \int_{\Omega} \mathrm{d}^4x \int_{\Omega} \mathrm{d}^4y \, \frac{\delta \mathcal{L}}{\delta \phi(y)} \frac{\mathrm{d}\phi'(y)}{\mathrm{d}a}\Bigg|_{a=0} \\ & + \int_{\Omega} \mathrm{d}^4x \, \mathcal{L} \frac{\mathrm{d}}{\mathrm{d}a} \mathrm{Det}\left[\frac{\partial x'}{\partial x}\right]\Bigg|_{a=0} \end{aligned}$$

$$+\int_\Omega \mathrm{d}^4x\,\frac{\mathrm{d}\mathcal{L}}{\mathrm{d}x^\alpha}\frac{\mathrm{d}x'^\alpha}{\mathrm{d}a}\bigg|_{a=0} \tag{C.6}$$

Suppose the lagrangian has at most second derivatives,

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}a}S'_a[\phi']\bigg|_{a=0} &= \int_\Omega \mathrm{d}^4x \int_\Omega \mathrm{d}^4y \left[\frac{\partial\mathcal{L}}{\partial\phi(x)}\delta^{(4)}(x-y) + \frac{\partial\mathcal{L}}{\partial\partial_{x\mu}\phi(x)}\partial_{x\mu}\delta^{(4)}(x-y)\right]\frac{\mathrm{d}\phi'(y)}{\mathrm{d}a}\bigg|_{a=0} \\ &\quad + \int_\Omega \mathrm{d}^4x\,\mathcal{L}\frac{\mathrm{d}}{\mathrm{d}a}\mathrm{Det}\left[\frac{\partial x'}{\partial x}\right]\bigg|_{a=0} \\ &\quad + \int_\Omega \mathrm{d}^4x\,\frac{\mathrm{d}\mathcal{L}}{\mathrm{d}x^\alpha}\frac{\mathrm{d}x'^\alpha}{\mathrm{d}a}\bigg|_{a=0}\end{aligned} \tag{C.7}$$

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}a}S'_a[\phi']\bigg|_{a=0} &= \int_\Omega \mathrm{d}^4x \left[\frac{\partial\mathcal{L}}{\partial\phi(x)}\frac{\mathrm{d}\phi'(x)}{\mathrm{d}a}\bigg|_{a=0} + \frac{\partial\mathcal{L}}{\partial\partial_\mu\phi(x)}\partial_\mu\frac{\mathrm{d}\phi'(x)}{\mathrm{d}a}\bigg|_{a=0}\right] \\ &\quad + \int_\Omega \mathrm{d}^4x\,\mathcal{L}\frac{\mathrm{d}}{\mathrm{d}a}\mathrm{Det}\left[\frac{\partial x'}{\partial x}\right]\bigg|_{a=0} \\ &\quad + \int_\Omega \mathrm{d}^4x\,\frac{\mathrm{d}\mathcal{L}}{\mathrm{d}x^\alpha}\frac{\mathrm{d}x'^\alpha}{\mathrm{d}a}\bigg|_{a=0}\end{aligned} \tag{C.8}$$

Now impose the equations of motion,

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}a}S'_a[\phi']\bigg|_{a=0} &= \int_\Omega \mathrm{d}^4x \left[\partial_\mu\frac{\partial\mathcal{L}}{\partial\partial_\mu\phi(x)}\frac{\mathrm{d}\phi'(x)}{\mathrm{d}a}\bigg|_{a=0} + \frac{\partial\mathcal{L}}{\partial\partial_\mu\phi(x)}\partial_\mu\frac{\mathrm{d}\phi'(x)}{\mathrm{d}a}\bigg|_{a=0}\right] \\ &\quad + \int_\Omega \mathrm{d}^4x\,\mathcal{L}\frac{\mathrm{d}}{\mathrm{d}a}\mathrm{Det}\left[\frac{\partial x'}{\partial x}\right]\bigg|_{a=0} \\ &\quad + \int_\Omega \mathrm{d}^4x\,\frac{\mathrm{d}\mathcal{L}}{\mathrm{d}x^\alpha}\frac{\mathrm{d}x'^\alpha}{\mathrm{d}a}\bigg|_{a=0}\end{aligned} \tag{C.9}$$

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}a}S'_a[\phi']\bigg|_{a=0} &= \int_\Omega \mathrm{d}^4x\,\partial_\mu\left[\frac{\partial\mathcal{L}}{\partial\partial_\mu\phi(x)}\frac{\mathrm{d}\phi'(x)}{\mathrm{d}a}\bigg|_{a=0}\right] \\ &\quad + \int_\Omega \mathrm{d}^4x\,\mathcal{L}\frac{\mathrm{d}}{\mathrm{d}a}\mathrm{Det}\left[\frac{\partial x'}{\partial x}\right]\bigg|_{a=0} \\ &\quad + \int_\Omega \mathrm{d}^4x\,\frac{\mathrm{d}\mathcal{L}}{\mathrm{d}x^\alpha}\frac{\mathrm{d}x'^\alpha}{\mathrm{d}a}\bigg|_{a=0}\end{aligned} \tag{C.10}$$

The last missing piece is to compute the determinant,

$$\mathrm{Det}\left[\frac{\partial x'}{\partial x}\right] = -\frac{1}{4!}\epsilon_{\alpha\beta\gamma\delta}\epsilon^{\mu\nu\rho\sigma}\frac{\partial x'^\alpha}{\partial x^\mu}\frac{\partial x'^\beta}{\partial x^\nu}\frac{\partial x'^\gamma}{\partial x^\rho}\frac{\partial x'^\delta}{\partial x^\sigma} \tag{C.11}$$

$$\frac{\mathrm{d}}{\mathrm{d}a}\left[\mathrm{Det}\left[\frac{\partial x'}{\partial x}\right]\right]\bigg|_{a=0} = -\frac{1}{4!}\epsilon_{\alpha\beta\gamma\delta}\epsilon^{\mu\nu\rho\sigma}\frac{\mathrm{d}}{\mathrm{d}a}\left[\frac{\partial x'^\alpha}{\partial x^\mu}\frac{\partial x'^\beta}{\partial x^\nu}\frac{\partial x'^\gamma}{\partial x^\rho}\frac{\partial x'^\delta}{\partial x^\sigma}\right]\bigg|_{a=0} \tag{C.12}$$

$$\frac{\mathrm{d}}{\mathrm{d}a}\left[\mathrm{Det}\left[\frac{\partial x'}{\partial x}\right]\right]\bigg|_{a=0} = -\frac{1}{3!}\epsilon_{\alpha\beta\gamma\delta}\epsilon^{\mu\nu\rho\sigma}\frac{\mathrm{d}}{\mathrm{d}a}\left[\frac{\partial x'^{\alpha}}{\partial x^{\mu}}\right]\bigg|_{a=0}\frac{\partial x'^{\beta}}{\partial x^{\nu}}\bigg|_{a=0}\frac{\partial x'^{\gamma}}{\partial x^{\rho}}\bigg|_{a=0}\frac{\partial x'^{\delta}}{\partial x^{\sigma}}\bigg|_{a=0} \tag{C.13}$$

$$\frac{\mathrm{d}}{\mathrm{d}a}\left[\mathrm{Det}\left[\frac{\partial x'}{\partial x}\right]\right]\bigg|_{a=0} = -\frac{1}{3!}\epsilon_{\alpha\beta\gamma\delta}\epsilon^{\mu\nu\rho\sigma}\frac{\mathrm{d}}{\mathrm{d}a}\left[\frac{\partial x'^{\alpha}}{\partial x^{\mu}}\right]\bigg|_{a=0} g^{\beta}{}_{\nu}g^{\gamma}{}_{\rho}g^{\delta}{}_{\sigma} \tag{C.14}$$

$$\frac{\mathrm{d}}{\mathrm{d}a}\left[\mathrm{Det}\left[\frac{\partial x'}{\partial x}\right]\right]\bigg|_{a=0} = -\frac{1}{3!}\epsilon_{\alpha\nu\rho\sigma}\epsilon^{\mu\nu\rho\sigma}\frac{\mathrm{d}}{\mathrm{d}a}\left[\frac{\partial x'^{\alpha}}{\partial x^{\mu}}\right]\bigg|_{a=0} \tag{C.15}$$

$$\frac{\mathrm{d}}{\mathrm{d}a}\left[\mathrm{Det}\left[\frac{\partial x'}{\partial x}\right]\right]\bigg|_{a=0} = \frac{1}{3!}3!g_{\alpha}{}^{\mu}\frac{\mathrm{d}}{\mathrm{d}a}\left[\frac{\partial x'^{\alpha}}{\partial x^{\mu}}\right]\bigg|_{a=0} \tag{C.16}$$

$$\frac{\mathrm{d}}{\mathrm{d}a}\left[\mathrm{Det}\left[\frac{\partial x'}{\partial x}\right]\right]\bigg|_{a=0} = \frac{\mathrm{d}}{\mathrm{d}a}\left[\frac{\partial x'^{\alpha}}{\partial x^{\alpha}}\right]\bigg|_{a=0} = \frac{\mathrm{d}}{\mathrm{d}x^{\alpha}}\frac{\mathrm{d}x'^{\alpha}}{\mathrm{d}a}\bigg|_{a=0} \tag{C.17}$$

Thus,

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}a}S'_a[\phi']\bigg|_{a=0} &= \int_{\Omega}\mathrm{d}^4x\,\partial_{\mu}\left[\frac{\partial\mathcal{L}}{\partial\partial_{\mu}\phi(x)}\frac{\mathrm{d}\phi'(x)}{\mathrm{d}a}\bigg|_{a=0}\right] \\ &\quad + \int_{\Omega}\mathrm{d}^4x\,\mathcal{L}\frac{\mathrm{d}}{\mathrm{d}x^{\alpha}}\frac{\mathrm{d}x'^{\alpha}}{\mathrm{d}a}\bigg|_{a=0} \\ &\quad + \int_{\Omega}\mathrm{d}^4x\,\frac{\mathrm{d}\mathcal{L}}{\mathrm{d}x^{\alpha}}\frac{\mathrm{d}x'^{\alpha}}{\mathrm{d}a}\bigg|_{a=0}\end{aligned} \tag{C.18}$$

$$\frac{\mathrm{d}}{\mathrm{d}a}S'_a[\phi']\bigg|_{a=0} = \int_{\Omega}\mathrm{d}^4x\,\frac{\mathrm{d}}{\mathrm{d}x^{\mu}}\left[\frac{\partial\mathcal{L}}{\partial\partial_{\mu}\phi(x)}\frac{\mathrm{d}\phi'(x)}{\mathrm{d}a}\bigg|_{a=0} + \mathcal{L}\frac{\mathrm{d}x'^{\mu}}{\mathrm{d}a}\bigg|_{a=0}\right] \tag{C.19}$$

It's trivial to include indicies,

$$\frac{\mathrm{d}}{\mathrm{d}a^I}S'_a[\phi']\bigg|_{a^I=0} = \int_{\Omega}\mathrm{d}^4x\,\frac{\mathrm{d}}{\mathrm{d}x^{\mu}}\left[\frac{\partial\mathcal{L}}{\partial\partial_{\mu}\phi(x)}\frac{\mathrm{d}\phi'(x)}{\mathrm{d}a^I}\bigg|_{a^I=0} + \mathcal{L}\frac{\mathrm{d}x'^{\mu}}{\mathrm{d}a^I}\bigg|_{a^I=0}\right] \tag{C.20}$$

So that the conserved current is,

$$J^{\mu}{}_{I} = \frac{\partial\mathcal{L}}{\partial\partial_{\mu}\phi(x)}\frac{\mathrm{d}\phi'(x)}{\mathrm{d}a^I}\bigg|_{a^I=0} + \mathcal{L}\frac{\mathrm{d}x'^{\mu}}{\mathrm{d}a^I}\bigg|_{a^I=0} \tag{C.21}$$

D Noether's Second Theorem

This is an extension of the last result which deals with local transformations that leave the action invariant. Notice, we aren't saying symmetry, but invariance, this will be clear in the end of this appendix. Let the transformation be parametrized by a function $a(x)$,

$$\begin{cases} x^\mu & \to x'^\mu = x'^\mu(x; a(x)) \\ \phi(x) & \to \phi'(x') = \phi'(\phi(x), x; a(x)) \end{cases} \Rightarrow S[\phi] = S'[\phi'; a] \tag{D.1}$$

The reasoning proceeds as before, but now the derivatives are functional ones, also, due to distributional matters, is of need to include a test function,

$$\begin{aligned} \int_\Omega \mathrm{d}^4 z \, \frac{\delta S'[\phi'; a]}{\delta a(z)} \psi(z) = & \int_\Omega \mathrm{d}^4 z \, \psi(z) \int_\Omega \mathrm{d}^4 x \int_\Omega \mathrm{d}^4 y \, \frac{\delta \mathcal{L}}{\delta \phi(y)} \frac{\delta \phi'(y)}{\delta a(z)} \bigg|_{a \equiv 0} \\ & + \int_\Omega \mathrm{d}^4 z \, \psi(z) \int_\Omega \mathrm{d}^4 x \, \mathcal{L} \frac{\mathrm{d}}{\mathrm{d}x^\alpha} \frac{\delta x'^\alpha}{\delta a(z)} \bigg|_{a \equiv 0} \\ & + \int_\Omega \mathrm{d}^4 z \, \psi(z) \int_\Omega \mathrm{d}^4 x \, \frac{\mathrm{d}\mathcal{L}}{\mathrm{d}x^\alpha} \frac{\delta x'^\alpha}{\delta a(z)} \bigg|_{a \equiv 0} \end{aligned} \tag{D.2}$$

$$\begin{aligned} \int_\Omega \mathrm{d}^4 z \, \frac{\delta S'[\phi'; a]}{\delta a(z)} \psi(z) = & \int_\Omega \mathrm{d}^4 z \, \psi(z) \int_\Omega \mathrm{d}^4 x \int_\Omega \mathrm{d}^4 y \left[\frac{\partial \mathcal{L}}{\partial \phi(x)} \delta^{(4)}(x-y) + \frac{\partial \mathcal{L}}{\partial \partial_{x\mu} \phi(x)} \partial_{x\mu} \delta^{(4)}(x-y) \right] \frac{\delta \phi'(y)}{\delta a(z)} \bigg|_{a \equiv 0} \\ & + \int_\Omega \mathrm{d}^4 z \, \psi(z) \int_\Omega \mathrm{d}^4 x \, \frac{\mathrm{d}}{\mathrm{d}x^\alpha} \left[\mathcal{L} \frac{\delta x'^\alpha}{\delta a(z)} \bigg|_{a \equiv 0} \right] \end{aligned} \tag{D.3}$$

$$\begin{aligned} \int_\Omega \mathrm{d}^4 z \, \frac{\delta S'[\phi'; a]}{\delta a(z)} \psi(z) = & \int_\Omega \mathrm{d}^4 z \, \psi(z) \int_\Omega \mathrm{d}^4 x \int_\Omega \mathrm{d}^4 y \, \frac{\partial \mathcal{L}}{\partial \phi(x)} \delta^{(4)}(x-y) \frac{\delta \phi'(y)}{\delta a(z)} \bigg|_{a \equiv 0} \\ & + \int_\Omega \mathrm{d}^4 z \, \psi(z) \int_\Omega \mathrm{d}^4 x \int_\Omega \mathrm{d}^4 y \, \frac{\partial \mathcal{L}}{\partial \partial_{x\mu} \phi(x)} \partial_{x\mu} \delta^{(4)}(x-y) \frac{\delta \phi'(y)}{\delta a(z)} \bigg|_{a \equiv 0} \\ & + \int_\Omega \mathrm{d}^4 z \, \psi(z) \int_\Omega \mathrm{d}^4 x \, \frac{\mathrm{d}}{\mathrm{d}x^\alpha} \left[\mathcal{L} \frac{\delta x'^\alpha}{\delta a(z)} \bigg|_{a \equiv 0} \right] \end{aligned} \tag{D.4}$$

$$\begin{aligned} \int_\Omega \mathrm{d}^4 z \, \frac{\delta S'[\phi'; a]}{\delta a(z)} \psi(z) = & \int_\Omega \mathrm{d}^4 z \, \psi(z) \int_\Omega \mathrm{d}^4 x \, \frac{\partial \mathcal{L}}{\partial \phi(x)} \frac{\delta \phi'(x)}{\delta a(z)} \bigg|_{a \equiv 0} \\ & + \int_\Omega \mathrm{d}^4 z \, \psi(z) \int_\Omega \mathrm{d}^4 x \int_\Omega \mathrm{d}^4 y \, \partial_{x\mu} \left[\frac{\partial \mathcal{L}}{\partial \partial_\mu \phi(x)} \delta^{(4)}(x-y) \right] \frac{\delta \phi'(y)}{\delta a(z)} \bigg|_{a \equiv 0} \\ & - \int_\Omega \mathrm{d}^4 z \, \psi(z) \int_\Omega \mathrm{d}^4 x \int_\Omega \mathrm{d}^4 y \, \partial_{x\mu} \frac{\partial \mathcal{L}}{\partial \partial_\mu \phi(x)} \delta^{(4)}(x-y) \frac{\delta \phi'(y)}{\delta a(z)} \bigg|_{a \equiv 0} \\ & + \int_\Omega \mathrm{d}^4 z \, \psi(z) \int_\Omega \mathrm{d}^4 x \, \frac{\mathrm{d}}{\mathrm{d}x^\alpha} \left[\mathcal{L} \frac{\delta x'^\alpha}{\delta a(z)} \bigg|_{a \equiv 0} \right] \end{aligned} \tag{D.5}$$

$$\begin{aligned} \int_\Omega \mathrm{d}^4 z \, \frac{\delta S'[\phi'; a]}{\delta a(z)} \psi(z) = & \int_\Omega \mathrm{d}^4 z \, \psi(z) \int_\Omega \mathrm{d}^4 x \, E[\phi(x); x] \frac{\delta \phi'(x)}{\delta a(z)} \bigg|_{a \equiv 0} \\ & + \int_\Omega \mathrm{d}^4 z \, \psi(z) \int_\Omega \mathrm{d}^4 x \, \partial_{x\mu} \left[\frac{\partial \mathcal{L}}{\partial \partial_\mu \phi(x)} \frac{\delta \phi'(x)}{\delta a(z)} \bigg|_{a \equiv 0} \right] \end{aligned}$$

$$+\int_\Omega \mathrm{d}^4z\,\psi(z)\int_\Omega \mathrm{d}^4x\,\frac{\mathrm{d}}{\mathrm{d}x^\alpha}\left[\mathcal{L}\frac{\delta x'^\alpha}{\delta a(z)}\bigg|_{a\equiv 0}\right] \tag{D.6}$$

Usually the space-time isn't transformed in a local transformation,

$$\begin{aligned}0 = &\int_\Omega \mathrm{d}^4z\,\psi(z)\int_\Omega \mathrm{d}^4x\,E[\phi(x);x]\frac{\delta\phi'(x)}{\delta a(z)}\bigg|_{a\equiv 0}\\ &+\int_\Omega \mathrm{d}^4z\,\psi(z)\int_{\partial\Omega} \mathrm{d}^3S_{x\mu}\,\frac{\partial\mathcal{L}}{\partial\partial_\mu\phi(x)}\frac{\delta\phi'(x)}{\delta a(z)}\bigg|_{a\equiv 0}\end{aligned} \tag{D.7}$$

As an example, let's use the electromagnetic case,

$$\begin{aligned}0 = &\int_\Omega \mathrm{d}^4z\,\psi(z)\int_\Omega \mathrm{d}^4x\,E^\nu\frac{\delta A'_\nu(x)}{\delta a(z)}\bigg|_{a\equiv 0}\\ &+\int_\Omega \mathrm{d}^4z\,\psi(z)\int_{\partial\Omega} \mathrm{d}^3S_{x\mu}\,\frac{\partial\mathcal{L}}{\partial\partial_\mu A_\nu(x)}\frac{\delta A'_\nu(x)}{\delta a(z)}\bigg|_{a\equiv 0}\end{aligned} \tag{D.8}$$

$$\begin{aligned}0 = &\int_\Omega \mathrm{d}^4z\,\psi(z)\int_\Omega \mathrm{d}^4x\,E^\nu\partial_{x\nu}\delta^{(4)}(x-z)\\ &+\int_\Omega \mathrm{d}^4z\,\psi(z)\int_{\partial\Omega} \mathrm{d}^3S_{x\mu}\,\frac{\partial\mathcal{L}}{\partial\partial_\mu A_\nu(x)}\partial_{x\nu}\delta^{(4)}(x-z)\end{aligned} \tag{D.9}$$

$$\begin{aligned}0 = &-\int_\Omega \mathrm{d}^4z\,\psi(z)\int_\Omega \mathrm{d}^4x\,\partial_{x\nu}E^\nu\delta^{(4)}(x-z)\\ &+\int_\Omega \mathrm{d}^4z\,\psi(z)\int_{\partial\Omega} \mathrm{d}^3S_{x\nu}\,E^\nu\delta^{(4)}(x-z)\\ &-\int_\Omega \mathrm{d}^4z\,\psi(z)\int_{\partial\Omega} \mathrm{d}^3S_{x\mu}\,\partial_{x\nu}\frac{\partial\mathcal{L}}{\partial\partial_\mu A_\nu(x)}\delta^{(4)}(x-z)\end{aligned} \tag{D.10}$$

$$0 = -\int_\Omega \mathrm{d}^4x\,\partial_\mu E^\mu\psi(x) + \int_{\partial\Omega} \mathrm{d}^3S_\mu\left[E^\mu - \partial_\nu\frac{\partial\mathcal{L}}{\partial\partial_\mu A_\nu(x)}\right]\psi(x) \tag{D.11}$$

This is all without imposing the equations of motion,

E Boundary Conditions

Let our spacetime be a four dimensional manifold Ω, with boundary $\partial\Omega = \Sigma_- \cup \Sigma_+ \cup \Gamma$. Our action principle is given by,

$$S[\phi] = \int\limits_{\Omega} \mathrm{d}^4x\, \mathcal{L}[\phi(x), x] + \int\limits_{\partial\Omega} \mathrm{d}^3S_{x\alpha}\, \ell^\alpha[\phi(x), x] \tag{E.1}$$

The variation of the action cannot have contributions from the bulk of Γ, and also is done with compact support outside of $\Sigma_- \cup \Sigma_+$, that is, on $\tilde{\Omega} = \Omega' \cup \Gamma'$,

$$\int\limits_{\tilde{\Omega}} \mathrm{d}^4y\, \frac{\delta S[\phi]}{\delta\phi(y)}\psi(y) = \int\limits_{\tilde{\Omega}} \mathrm{d}^4y \int\limits_{\Omega} \mathrm{d}^4x\, \frac{\delta\mathcal{L}[x]}{\delta\phi(y)}\psi(y) + \int\limits_{\tilde{\Omega}} \mathrm{d}^4y \int\limits_{\partial\Omega} \mathrm{d}^3S_{x\alpha}\, \frac{\delta\ell^\alpha[x]}{\delta\phi(y)}\psi(y) \tag{E.2}$$

$$\begin{aligned}\int\limits_{\tilde{\Omega}} \mathrm{d}^4y\, \frac{\delta S[\phi]}{\delta\phi(y)}\psi(y) = &\int\limits_{\tilde{\Omega}} \mathrm{d}^4y \int\limits_{\Omega} \mathrm{d}^4x \left[\frac{\partial\mathcal{L}[x]}{\partial\phi(x)}\delta^{(4)}(x-y) + \frac{\partial\mathcal{L}[x]}{\partial\partial_\mu\phi(x)}\partial_\mu\delta^{(4)}(x-y)\right]\psi(y) \\ &+ \int\limits_{\tilde{\Omega}} \mathrm{d}^4y \int\limits_{\partial\Omega} \mathrm{d}^3S_{x\alpha}\, \frac{\delta\ell^\alpha[x]}{\delta\phi(y)}\psi(y)\end{aligned} \tag{E.3}$$

$$\begin{aligned}\int\limits_{\tilde{\Omega}} \mathrm{d}^4y\, \frac{\delta S[\phi]}{\delta\phi(y)}\psi(y) = &\int\limits_{\tilde{\Omega}} \mathrm{d}^4y \int\limits_{\Omega} \mathrm{d}^4x \left[\frac{\partial\mathcal{L}[x]}{\partial\phi(x)} - \partial_\mu\frac{\partial\mathcal{L}[x]}{\partial\partial_\mu\phi(x)}\right]\delta^{(4)}(x-y)\psi(y) \\ &+ \int\limits_{\tilde{\Omega}} \mathrm{d}^4y \int\limits_{\Omega} \mathrm{d}^4x\, \partial_{x\mu}\left[\frac{\partial\mathcal{L}[x]}{\partial\partial_\mu\phi(x)}\delta^{(4)}(x-y)\right]\psi(y) + \int\limits_{\tilde{\Omega}} \mathrm{d}^4y \int\limits_{\partial\Omega} \mathrm{d}^3S_{x\alpha}\, \frac{\delta\ell^\alpha[x]}{\delta\phi(y)}\psi(y)\end{aligned} \tag{E.4}$$

$$\begin{aligned}\int\limits_{\tilde{\Omega}} \mathrm{d}^4y\, \frac{\delta S[\phi]}{\delta\phi(y)}\psi(y) = &\int\limits_{\tilde{\Omega}} \mathrm{d}^4y \left[\frac{\partial\mathcal{L}[y]}{\partial\phi(y)} - \partial_\mu\frac{\partial\mathcal{L}[y]}{\partial\partial_\mu\phi(y)}\right]\psi(y) \\ &+ \int\limits_{\tilde{\Omega}} \mathrm{d}^4y \int\limits_{\partial\Omega} \mathrm{d}^3S_{x\alpha}\, \frac{\partial\mathcal{L}[x]}{\partial\partial_\alpha\phi(x)}\delta^{(4)}(x-y)\psi(y) + \int\limits_{\tilde{\Omega}} \mathrm{d}^4y \int\limits_{\partial\Omega} \mathrm{d}^3S_{x\alpha}\, \frac{\delta\ell^\alpha[x]}{\delta\phi(y)}\psi(y)\end{aligned} \tag{E.5}$$

$$\begin{aligned}\int\limits_{\tilde{\Omega}} \mathrm{d}^4y\, \frac{\delta S[\phi]}{\delta\phi(y)}\psi(y) = &\int\limits_{\tilde{\Omega}} \mathrm{d}^4y \left[\frac{\partial\mathcal{L}[y]}{\partial\phi(y)} - \partial_\mu\frac{\partial\mathcal{L}[y]}{\partial\partial_\mu\phi(y)}\right]\psi(y) \\ &+ \int\limits_{\partial\Omega} \mathrm{d}^3S_{x\alpha}\, \frac{\partial\mathcal{L}[x]}{\partial\partial_\alpha\phi(x)}\psi(x) + \int\limits_{\tilde{\Omega}} \mathrm{d}^4y \int\limits_{\partial\Omega} \mathrm{d}^3S_{x\alpha}\, \frac{\delta\ell^\alpha[x]}{\delta\phi(y)}\psi(y)\end{aligned} \tag{E.6}$$

For many cases $\ell^\alpha \equiv 0$ due to dynamical and covariance reasons. Of course ψ is zero on $\Sigma_\pm$, then, the equations of motion impose:

$$0 = \frac{\partial\mathcal{L}[\phi(x); x]}{\partial\phi(x)} - \partial_\mu\frac{\partial\mathcal{L}[\phi(x); x]}{\partial\partial_\mu\phi(x)}\bigg|_{x\in\Omega} \tag{E.7}$$

$$0 = n_\alpha(x)\partial^\alpha\phi(x)\bigg|_{x\in\Gamma} \tag{E.8}$$

With n_α being the normal direction in Γ. If we try to compute the conserved charge by Noether's theorem,

$$0 = \int\limits_{\partial\Omega} \mathrm{d}^3 S_\mu \left[\frac{\partial \mathcal{L}}{\partial \partial_\mu \phi(x)} \frac{\mathrm{d}\phi'(x)}{\mathrm{d}a} \bigg|_{a=0} + \mathcal{L} \frac{\mathrm{d}x'^\mu}{\mathrm{d}a} \bigg|_{a=0} \right] \tag{E.9}$$

$$0 = \int\limits_{\Sigma_+} \mathrm{d}^3 S_\mu \, J^\mu - \int\limits_{\Sigma_-} \mathrm{d}^3 S_\mu \, J^\mu + \int\limits_{\Gamma} \mathrm{d}^3 S_\mu \left[\frac{\partial \mathcal{L}}{\partial \partial_\mu \phi(x)} \frac{\mathrm{d}\phi'(x)}{\mathrm{d}a} \bigg|_{a=0} + \mathcal{L} \frac{\mathrm{d}x'^\mu}{\mathrm{d}a} \bigg|_{a=0} \right] \tag{E.10}$$

$$0 = \int\limits_{\Sigma_+} \mathrm{d}^3 S_\mu \, J^\mu - \int\limits_{\Sigma_-} \mathrm{d}^3 S_\mu \, J^\mu + \int\limits_{\Gamma} \mathrm{d}^3 S_\mu \, \mathcal{L} \frac{\mathrm{d}x'^\mu}{\mathrm{d}a} \bigg|_{a=0} \tag{E.11}$$

Now, if we have $\mathcal{L}\big|_{x\in\Gamma} = 0$ or $\frac{\mathrm{d}x'}{\mathrm{d}a}\big|_{a=0,x\in\Gamma} = 0$ then we get a conserved charge!

$$\int\limits_{\Sigma_+} \mathrm{d}^3 S_\mu \, J^\mu = \int\limits_{\Sigma_-} \mathrm{d}^3 S_\mu \, J^\mu \tag{E.12}$$

Now let's consider something other, let be a variation with change the values of the fields at the time-like boundaries, that is, a variation between different solutions,

$$\begin{aligned} \int\limits_{\Omega} \mathrm{d}^4 y \, \frac{\delta S[\phi]}{\delta \phi(y)} \psi_1(y) = & \int\limits_{\Omega} \mathrm{d}^4 y \, E[\phi(y); y] \psi_1(y) \\ & + \int\limits_{\partial\Omega} \mathrm{d}^3 S_\alpha \, \Theta^\alpha[\phi(x); x] \psi_1(x) \end{aligned} \tag{E.13}$$

$$\begin{aligned} \int\limits_{\Omega} \mathrm{d}^4 y \int\limits_{\Omega} \mathrm{d}^4 z \, \frac{\delta^2 S[\phi]}{\delta\phi(z)\delta\phi(y)} \psi_1(y)\psi_2(z) = & \int\limits_{\Omega} \mathrm{d}^4 y \int\limits_{\Omega} \mathrm{d}^4 z \, \frac{\delta E[\phi(y); y]}{\delta\phi(z)} \psi_1(y)\psi_2(z) \\ & + \int\limits_{\partial\Omega} \mathrm{d}^3 S_\alpha \int\limits_{\Omega} \mathrm{d}^4 z \, \frac{\delta \Theta^\alpha[\phi(x); x]}{\delta\phi(z)} \psi_1(x)\psi_2(z) \end{aligned} \tag{E.14}$$

$$\tag{E.15}$$

Antisymmetrize in $1 \leftrightarrow 2$,

$$\begin{aligned} 0 = & \int\limits_{\Omega} \mathrm{d}^4 y \int\limits_{\Omega} \mathrm{d}^4 z \left(\frac{\delta E[\phi(y); y]}{\delta\phi(z)} - \frac{\delta E[\phi(z); z]}{\delta\phi(y)} \right) \psi_1(y)\psi_2(z) \\ & + \int\limits_{\partial\Omega} \mathrm{d}^3 S_\alpha \int\limits_{\Omega} \mathrm{d}^4 z \left(\frac{\delta \Theta^\alpha[\phi(x); x]}{\delta\phi(z)} - \frac{\delta \Theta^\alpha[\phi(z); z]}{\delta\phi(x)} \right) \psi_1(x)\psi_2(z) \end{aligned} \tag{E.16}$$

As long as the equation of motion have just even order of derivatives with constant coefficients,

$$0 = \int\limits_{\partial\Omega} \mathrm{d}^3 S_\alpha \int\limits_{\Omega} \mathrm{d}^4 z \left(\partial_x^\alpha \delta^{(4)}(x-z) - \partial_z^\alpha \delta^{(4)}(z-x) \right) \psi_1(x)\psi_2(z) \tag{E.17}$$

$$0 = \int\limits_{\partial\Omega} \mathrm{d}^3 S_\alpha \left(\psi_1(x) \partial^\alpha \psi_2(x) - \psi_2(x) \partial^\alpha \psi_1(x) \right) \tag{E.18}$$

Suppose that for a given solution of the equations of motion ϕ, it's true that $\phi + \psi_i$ are also solutions of the equations of motion, then,

$$\int_{\Sigma_+} \mathrm{d}^3 S_\alpha \left(\psi_1(x)\partial^\alpha \psi_2(x) - \psi_2(x)\partial^\alpha \psi_1(x)\right) = \int_{\Sigma_-} \mathrm{d}^3 S_\alpha \left(\psi_1(x)\partial^\alpha \psi_2(x) - \psi_2(x)\partial^\alpha \psi_1(x)\right) \tag{E.19}$$

In other words, the functional,

$$\Omega(\psi_1, \psi_2) = \int_{\Sigma_t} \mathrm{d}^3\mathbf{x} \left(\psi_1 \dot{\psi}_2 - \psi_2 \dot{\psi}_1\right) \tag{E.20}$$

defined in the tangent space of the phase space of solutions, is independent of the time slice used to compute it. Taken into account that the phase space needs two variables for each degree of freedom, they are chosen as $\phi(\mathbf{x})$ and $\pi(\mathbf{x}) = \dot{\phi}(\mathbf{x})$, in this way, our functional can be rewritten as a phase space 2-form,

$$\Omega = \int_{\Sigma_t} \mathrm{d}^3\mathbf{x}\, \delta\phi(\mathbf{x}) \wedge \delta\pi(\mathbf{x}) \tag{E.21}$$

The main point is that this 2-form is non degenerated and can be inverted as,

$$\{\cdot, \cdot\} = \int_{\Sigma_t} \mathrm{d}^3\mathbf{x} \left(\frac{\delta}{\delta\phi(\mathbf{x})}\frac{\delta}{\delta\pi(\mathbf{x})} - \frac{\delta}{\delta\pi(\mathbf{x})}\frac{\delta}{\delta\phi(\mathbf{x})}\right) \tag{E.22}$$

Which provides the fundamental Poisson bracket,

$$\{\phi(t, \mathbf{x}), \pi(t, \mathbf{y})\} = \delta^{(3)}(\mathbf{x} - \mathbf{y}) \tag{E.23}$$

F Algebra of conserved quantities

F.A) Internal Symmetries

Given an internal symmetry of our action,

$$\begin{cases} \phi^I(x) & \to \phi'^I(x;a) \\ x^\mu & \to x'^\mu = x^\mu \end{cases} \Rightarrow S[\phi] = S'_a[\phi] \tag{F.1}$$

where I encode whatever index the fields of our theory may have. Suppose this symmetry is represented by some Lie Group/Algebra, that is, we have,

$$\phi'^I(x) = \exp\left(\mathrm{i}a_i T^i\right)^I{}_J \phi^J(x) \Rightarrow \left.\frac{\mathrm{d}\phi'^I(x)}{\mathrm{d}a^i}\right|_{a^i=0} = \mathrm{i}T^I{}_{Ji}\,\phi^J(x) \tag{F.2}$$

This transformation generates a conserved current,

$$J^\mu{}_i = \frac{\partial\mathcal{L}}{\partial\partial_\mu\phi^I(x)}\left.\frac{\mathrm{d}\phi'^I(x)}{\mathrm{d}a^i}\right|_{a^i=0} \tag{F.3}$$

$$J^\mu{}_i = -\partial^\mu\phi_I(x)\mathrm{i}T^I{}_{Ji}\,\phi^J(x) \tag{F.4}$$

$$J^0{}_i = -\pi_I\mathrm{i}T^I{}_{Ji}\,\phi^J \tag{F.5}$$

Compute the conserved charge,

$$Q_i = \int \mathrm{d}^3\mathbf{x}\, J^0{}_i = -\int \mathrm{d}^3\mathbf{x}\,\pi_I\mathrm{i}T^I{}_{Ji}\,\phi^J \tag{F.6}$$

Now, the Poisson bracket,

$$\{Q_i, Q_j\} = \int \mathrm{d}^3\mathbf{x}\left(\frac{\delta Q_i}{\delta\phi^I(\mathbf{x})}\frac{\delta Q_j}{\delta\pi_I(\mathbf{x})} - \frac{\delta Q_i}{\delta\pi_I(\mathbf{x})}\frac{\delta Q_j}{\delta\phi^I(\mathbf{x})}\right) \tag{F.7}$$

$$\{Q_i, Q_j\} = (-)^2\int \mathrm{d}^3\mathbf{x}\left(\pi_K\mathrm{i}T^K{}_{Ii}\,\mathrm{i}T^I{}_{Jj}\,\phi^J - \mathrm{i}T^I{}_{Ji}\,\phi^J\pi_K\mathrm{i}T^K{}_{Ij}\right) \tag{F.8}$$

$$\{Q_i, Q_j\} = \mathrm{i}^2\int \mathrm{d}^3\mathbf{x}\,\pi_K\left(T^K{}_{Ii}T^I{}_{Jj} - T^K{}_{Ij}T^I{}_{Ji}\right)\phi^J \tag{F.9}$$

$$\{Q_i, Q_j\} = \mathrm{i}^2\int \mathrm{d}^3\mathbf{x}\,\pi_K\Big([T_i, T_j]^K{}_J\Big)\phi^J \tag{F.10}$$

Now, as $T^K{}_{Ii}$ are the generators of a Lie Algebra, they obey an commutation relation of the form, $[T_i, T_j] = \mathrm{i}f_{ij}{}^k T_k$, thus,

$$\{Q_i, Q_j\} = \mathrm{i}^2\int \mathrm{d}^3\mathbf{x}\,\pi_K\left(\mathrm{i}f_{ij}{}^k T^K{}_{Jk}\right)\phi^J \tag{F.11}$$

$$\{Q_i, Q_j\} = -\mathrm{i}^2 f_{ij}{}^k(-)\int \mathrm{d}^3\mathbf{x}\,\pi_K\mathrm{i}T^K{}_{Jk}\,\phi^J \tag{F.12}$$

$$\{Q_i, Q_j\} = f_{ij}{}^k Q_k \tag{F.13}$$

In the quantum theory, at least from the canonical quantization POV, we have the map $\left[\hat{A}, \hat{B}\right] = \mathrm{i}\hbar\{A, B\}$, therefore, each classical conserved charge should be correspondent to a conserved charge operator $\hat{Q}_i$ which *should* satisfy the same Lie Algebra,

$$\left[\hat{Q}_i, \hat{Q}_j\right] = \mathrm{i}\hbar f_{ij}{}^k \hat{Q}_k \tag{F.14}$$

Almost always this isn't the case, and the symmetry has a anomaly in the quantum theory, that is, a transformation which is a symmetry in the classical theory isn't anymore a symmetry when the theory is quantized. For these cases, the most common and physical relevant way an anomaly shows is through a central extension of the Lie Algebra,

$$\left[\hat{Q}_i, \hat{Q}_j\right] = \mathrm{i}\hbar f_{ij}{}^k \hat{Q}_k + \hbar^2 C_{[ij]} \mathbb{1} \tag{F.15}$$

The name *central extension* is so because $C_{[ij]}$ are just numbers, that is, they commute with every other generator of the algebra, and thus, they belong to the center of the algebra — the maximum set of mutually commuting generators —. The $\hbar^2$ is needed for dimensional reasons (check it!), and it explain why this don't show up in the classical theory, if we consider the limit of,

$$\{Q_i, Q_f\} := \lim_{\hbar \to 0} \frac{1}{\mathrm{i}\hbar} \left[\hat{Q}_i, \hat{Q}_j\right] = f_{ij}{}^k Q_k \tag{F.16}$$

That is, the $\hbar^2$ term drop off, it's a true quantum effect. There are other ways of understanding why the central extensions of algebras show up in the process of quantization, there is a close relation with the need to consider the double cover of Lie Groups/Algebras when going from classical symmetries to quantum symmetries. Most of the anomalies are harmless, but, if there is an gauge anomaly the theory is physically inconsistent.

Luckily, the Poincaré Group/Algebra does not allow any kind of central extension (see Weinberg for proof), hence, the Poincaré symmetry is realized correctly in the quantum theory.

Other relevant property of the conserved charges is: they generate the symmetry transformation in the phase space variables,

$$\left\{Q_i, \phi^I(x)\right\} = \int \mathrm{d}^3\mathbf{y} \left(\frac{\delta Q_i}{\delta \phi^K(\mathbf{y})} \frac{\delta \phi^I(t, \mathbf{x})}{\delta \pi_K(\mathbf{y})} - \frac{\delta Q_i}{\delta \pi_K(\mathbf{y})} \frac{\delta \phi^I(t, \mathbf{x})}{\delta \phi^K(\mathbf{y})} \right) \tag{F.17}$$

$$\left\{Q_i, \phi^I(x)\right\} = (-)^2 \int \mathrm{d}^3\mathbf{y}\, \mathrm{i}T^K{}_{Ji}\, \phi^J \delta^I{}_K \delta^{(3)}(\mathbf{x} - \mathbf{y}) \tag{F.18}$$

$$\left\{Q_i, \phi^I(x)\right\} = \mathrm{i}T^I{}_{Ji}\, \phi^J \equiv \left.\frac{\mathrm{d}\phi'^I}{\mathrm{d}a^i}\right|_{a^I=0} \tag{F.19}$$

F.B) Space-Time Symmetries

Let's start with the easiest one, translations,

$$\begin{cases} \phi^I(x) & \to \phi'^I(x') = \phi^I(x) \\ x^\mu & \to x'^\mu = x^\mu + a^\mu \end{cases} \Rightarrow S[\phi] = S'_a[\phi] \tag{F.20}$$

Thus,

$$\left.\frac{\mathrm{d}\phi'^I(x)}{\mathrm{d}a^\nu}\right|_{a^\nu=0} = \mathrm{i}T^I{}_{J\nu}\, \phi^J(x) \tag{F.21}$$

$$\left.\frac{\mathrm{d}\phi^I(x-a)}{\mathrm{d}a^\nu}\right|_{a^\nu=0} = \mathrm{i}T^I{}_{J\nu}\,\phi^J(x) \tag{F.22}$$

$$-\partial_\nu\phi^I = \mathrm{i}T^I{}_{J\nu}\,\phi^J(x) \tag{F.23}$$

$$\mathrm{i}\mathrm{i}\delta^I{}_J\partial_\nu\phi^J = \mathrm{i}T^I{}_{J\nu}\,\phi^J(x) \tag{F.24}$$

And,

$$\left.\frac{\mathrm{d}x'^\mu}{\mathrm{d}a^\nu}\right|_{a^\nu=0} = g^\mu{}_\nu \tag{F.25}$$

And thus the conserved current is,

$$J^\mu{}_\nu = \frac{\partial\mathcal{L}}{\partial\partial_\mu\phi^I(x)}\left.\frac{\mathrm{d}\phi'^I(x)}{\mathrm{d}a^\nu}\right|_{a^\nu=0} + \mathcal{L}\left.\frac{\mathrm{d}x'^\mu}{\mathrm{d}a^\nu}\right|_{a^\nu=0} \tag{F.26}$$

$$J^\mu{}_\nu = -\frac{\partial\mathcal{L}}{\partial\partial_\mu\phi^I(x)}\partial_\nu\phi^I + \mathcal{L}g^\mu{}_\nu \tag{F.27}$$

$$J^\mu{}_\nu = \partial^\mu\phi_I\partial_\nu\phi^I + \mathcal{L}g^\mu{}_\nu \tag{F.28}$$

$$J^0{}_\nu = \pi_I\partial_\nu\phi^I + \mathcal{L}g^0{}_\nu \tag{F.29}$$

The momentum is,

$$P_\nu = \int \mathrm{d}^3\mathbf{x}\,J^0{}_\nu = \int \mathrm{d}^3\mathbf{x}\left(\pi_I\partial_\nu\phi^I + \mathcal{L}g^0{}_\nu\right) \tag{F.30}$$